



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
04-09-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. At $t = 0$, an electron (e, m) leaves one plate of a parallel plate capacitor with negligible initial velocity. An accelerating voltage, varying as $V = kt$, is applied between the plates. The separation between the plates is ℓ , velocity with which the electron reaches the opposite plate is :

(A) $\frac{ek}{m\ell} \left[\frac{3m\ell^2}{ek} \right]^{\frac{2}{3}}$

(B) $\frac{1}{2} \frac{ek}{m\ell} \left[\frac{6m\ell^2}{ek} \right]^{\frac{2}{3}}$

(C) $\frac{2ek}{m\ell} \left[\frac{6m\ell^2}{ek} \right]^{\frac{2}{3}}$

(D) $\frac{ek}{m\ell} \left[\frac{6m\ell^2}{ek} \right]^{\frac{1}{3}}$

2. The potential of a certain electric field in the 3-D space is given as $V = \alpha(xy - z^2)$, α is a constant. The projection of $\vec{E}$ on to the direction of vector $\vec{A} = \hat{i} + \hat{j} + 3\hat{k}$ at a point $(2, 1 - 3)$ is:

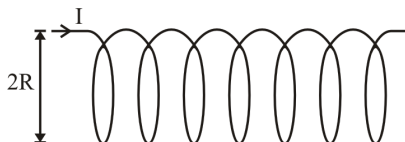
(A) Zero

(B) $\frac{-21\alpha}{\sqrt{11}}$

(C) $-\sqrt{11} \alpha$

(D) $\frac{1}{\sqrt{11}}$

3. The function $y(x, t) = (15.0 \text{ cm}) \cos (\pi x - 15 \pi t)$, with x in meters and t in seconds, describes a wave on a taut string. What is the transverse speed for a point on the string at an instant when that point has the displacement $y = +12.0 \text{ cm}$?
- (A) $180 \pi \text{ cm/s}$ (B) $225 \pi \text{ cm/s}$
 (C) $45 \pi \text{ cm/s}$ (D) $135 \pi \text{ cm/s}$
4. A current I flows in a long single layer solenoid with cross-sectional radius R . Number of turns per unit length of the solenoid is equal to n . Find the maximum possible value of current I that can be maintained without rupturing, if F_0 is the maximum tensile force the wire of solenoid can withstand.



- (A) $I_{\max} = \sqrt{\frac{F_0}{\mu_0 n R}}$
 (B) $I_{\max} = \sqrt{\frac{F_0}{2\mu_0 n R}}$
 (C) $I_{\max} = \frac{1}{2} \sqrt{\frac{F_0}{\mu_0 n R}}$
 (D) $I_{\max} = \sqrt{\frac{2F_0}{\mu_0 n R}}$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

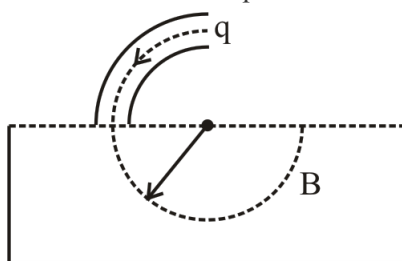
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. A charged particle flies through the electric field of a cylindrical capacitor and gets into a uniform transverse magnetic field with induction B as shown in figure. In the capacitor the particle moves along the arc of a circle, in the magnetic field, along a semicircle of radius ' r '. The potential difference applied to the capacitor is equal to V_0 , the radii of electrodes are equal to ' a ' and ' b ' with $a < b$



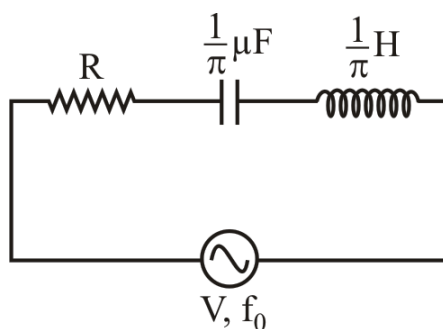
(A) The velocity of the particle is $v = \frac{V_0}{2Br \ln \left(\frac{b}{a} \right)}$

(B) The velocity of the particle is $v = \frac{V_0}{Br \ln \left(\frac{b}{a} \right)}$

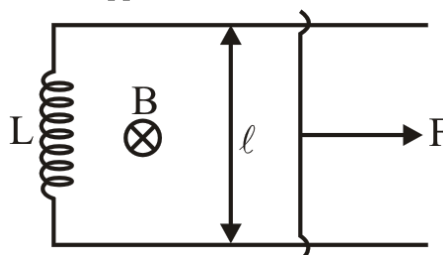
(C) The specific charge ratio $\frac{q}{m} = \frac{V_0}{B^2 r^2 \ln \left(\frac{b}{a} \right)}$

(D) The specific charge ratio $\frac{q}{m} = \frac{V_0}{2B^2 r^2 \ln \left(\frac{b}{a} \right)}$

6. For an AC circuit shown in figure, the supply voltage has constant rms value V . Circuit is in resonance at frequency, f_0 . At resonance

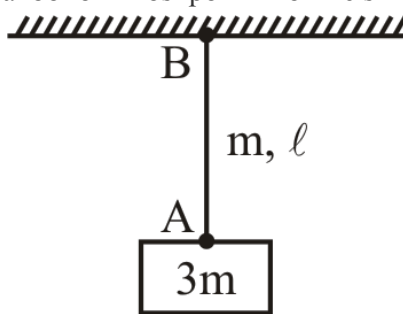


- (A) Current in circuit is $\frac{V}{R}$
- (B) $f_0 = 500$ Hz.
- (C) Voltage across the capacitor is 180° out of phase with the voltage across inductor.
- (D) Current in circuit is $\frac{V}{2R}$.
7. Figure shows conducting and smooth fixed rails in horizontal plane. An inductor is connected between rails. There exists a uniform constant vertical magnetic field B . Slider having mass m length ℓ was initially at rest. Now a constant force F is applied on slider as shown in figure. Then :

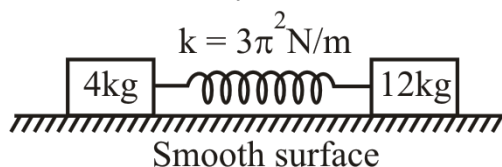


- (A) Angular frequency of slider would be $\frac{B\ell}{\sqrt{mL}}$.
- (B) Angular frequency of slider would be $\frac{B\ell}{mL}$.
- (C) Slider performs SHM
- (D) Slider moves with constant acceleration.
8. A rope, under a tension of 200 N and fixed at both ends, oscillates in a second-harmonic standing wave pattern. The displacement of the rope is given by $y = (0.10\text{m})(\sin \pi x/2) \sin 12 \pi t$, where $x = 0$ at one end of the rope, x is in meters, and t is in seconds.
- (A) The length of the rope is 6m.
- (B) Mass/length of the rope is $\frac{25}{72}$ kg/m.
- (C) There is an antinode at $x = 3\text{m}$.
- (D) The particle at $x = 1\text{m}$ & $x = 1.5\text{m}$ always oscillate in same phase.

9. A block of mass $3m$ is suspended vertically by a rope of mass m and length ℓ as shown in figure. A wave pulse of wavelength λ_0 is generated at bottom most point A of the string.



- (A) Wavelength of wave pulse at point B is $\sqrt{\frac{3}{4}} \lambda_0$.
- (B) Wavelength of wave pulse at point B is $\sqrt{\frac{4}{3}} \lambda_0$.
- (C) Time taken by wave pulse to reach point B is $(2 - \sqrt{3}) \sqrt{\frac{\ell}{g}}$.
- (D) Time taken by wave pulse to reach point B is $2(2 - \sqrt{3}) \sqrt{\frac{\ell}{g}}$.
10. System shown at rest is slightly compressed and released at $t = 0$. Separation between the blocks is maximum for first time at $t = t_1$ and relative velocity between the blocks is maximum at $t = t_2$, then



- (A) $t_1 = 1\text{s}$
- (B) $t_1 = \frac{1}{2}\text{s}$
- (C) $t_2 = 1\text{s}$
- (D) $t_2 = \frac{1}{2}\text{s}$

SECTION-II (i) : (Maximum Marks: 12)

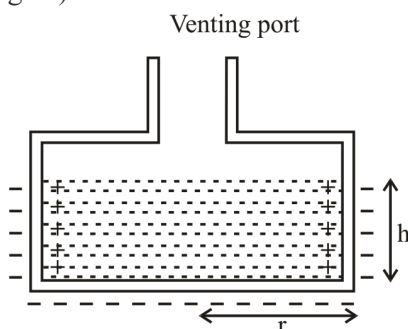
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A safety engineer needs to evaluate the practice of storing inflammable conducting liquids in non-conducting containers. The company supplying the liquid uses a cylindrical plastic container of radius $r = 20$ cm and fills it to a height of $h = 20$ cm, which is not equal to its full height. During handling, the exterior surface acquires a charge density of $\sigma = 2 \times 10^{-6} \text{ C/m}^2$, approximately uniform. This induces a charge separation within the liquid, because it is conducting (see figure).



1. How much charge is induced within the bulk of the liquid (in μC) Take : $\pi \approx 3$.
2. Assume that the capacitance of the central part relative to ground is nearly 36pF. Calculate the potential energy stored in the capacitor (in mJ).

Paragraph for Questions 3 and 4

An accumulator of emf 2V and negligible internal resistance is connected across a uniform wire of length 10 m and resistance 30Ω . The appropriate terminals of a cell of emf 1.5 V and internal resistance 1Ω is connected to one end of the wire and the other terminal of the cell connected through a sensitive galvanometer to a slider on the wire.

3. What is the length (in m) of the wire that will be required to produce zero deflection of the galvanometer.
4. What will be the balancing length (in m) when a coil of resistance 5Ω is placed in series with the accumulator.

Paragraph for Questions 5 and 6

A circuit takes a current of 3 A at a power factor of 0.6 lagging when connected to a 115 V- 50Hz supply. Another circuit takes a current of 5 A at a power factor of 0.707 leading when connected to the same supply. If the two circuits are connected in series across a 230 V, 50 Hz supply, then

5. What would be the value of current (in A) drawn by the circuit in ampere.
6. The power consumed by the circuit in kw.

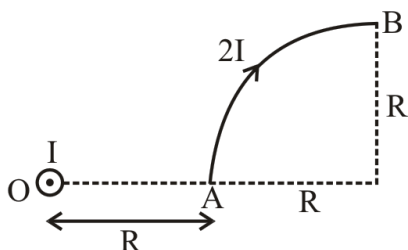
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

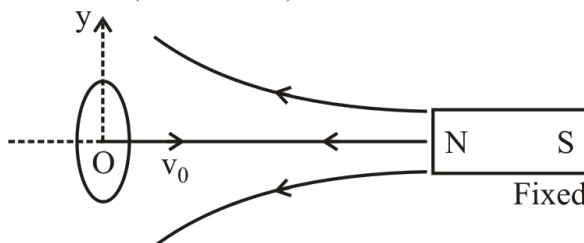
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. A point sound source has an output of $4\pi\mu\text{W}$. What is loudness level in (dB) at a distance of 10 m from the source ?
8. The magnetic force on current carrying wire AB due to an infinitely long current wire at O, is $\frac{\mu_0 I^2}{2\pi} \ln(N)$, find N :



9. A superconducting ring of mass 0.1 kg, inductance 0.02 H and radius 0.5 m is projected origin towards a magnet with velocity $v_0 = 1\text{m/s}$ as shown. An imaginary magnetic field along axis O varies as $B_x = B_0 (1 + x)$ and its radial component as $B_r = 2B_0$, where $B_0 = 0.1\text{ T}$ and x is in meters. Maximum distance traversed by ring (in cm) towards magnet is N cm, find N. (Take $\pi^2 = 10$)



PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

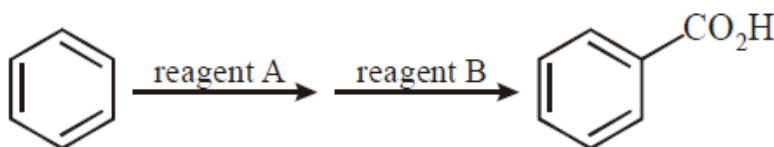
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

- Which of the following statement is incorrect?
 - In chromyl chloride test, formation of chromyl chloride is a non redox change
 - The brown ring test of NO_2^- can be done in presence of NO_3^- but brown ring test of NO_3^- cannot be done in presence of NO_2^-
 - $\text{CrO}_4^{2-}(\text{aq.})$ and $\text{Cr}_2\text{O}_7^{2-}(\text{aq.})$ can be inter changed by modifying the temperature
 - Borate can be detected by using CaF_2 along with conc. H_2SO_4 without touching the flame
- Which of the following observations do not confirm presence of NH_3 ?
 - Nessler's reagent gives brown ppt
 - Neutralising NaOH solution
 - Turning green solution of NiCl_2 to deep blue
 - Turning CuSO_4 to deep blue
- Which of the following reaction is non-redox?

(A) $\text{Cl}^- + \text{conc. H}_2\text{SO}_4 \rightarrow$	(B) $\text{Br}^- + \text{conc. H}_2\text{SO}_4 \rightarrow$
(C) $\text{Fe}^{3+} + \text{S}_2\text{O}_3^{2-} \xrightarrow[\text{standing}]{\text{on long}}$	(D) $\text{Cu}^{2+} + \text{S}_2\text{O}_3^{2-} \rightarrow$
- What reagents (conditions) are needed for the following reaction ?



- reagent A : $\text{CH}_3\text{COCl}/\text{AlCl}_3$; reagent B : Sn, HCl, heat
- reagent A : $\text{CH}_3\text{CH}_2\text{Cl}/\text{AlCl}_3$; reagent B : $\text{KMnO}_4, \text{heat}$
- reagent A : $\text{CH}_3\text{COCl}/\text{AlCl}_3$; reagent B : $\text{H}_2\text{NNH}_2, \text{KOH, H}_2\text{O, heat}$
- reagent A : $\text{HNO}_3, \text{H}_2\text{SO}_4$; reagent B : Sn, HCl, heat

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

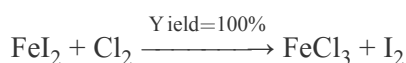
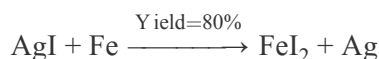
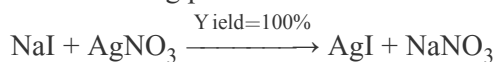
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

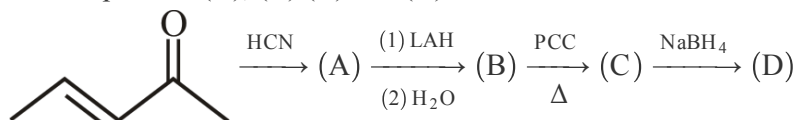
5. The following process has been used to obtain iodine from oil-field brines in California :

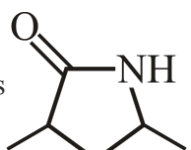
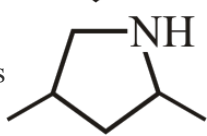
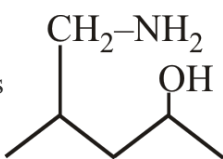
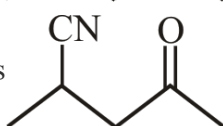


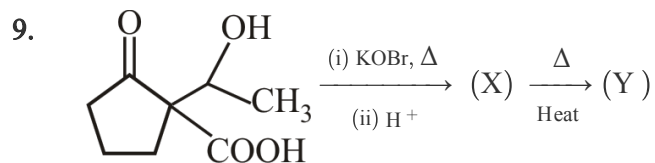
$\frac{1}{2 \times 10^4}$ part of the sample of I_2 produced is dissolved in water to make 1 litre solution. 10 ml of this solution requires 20 ml 0.1N hypo solution to decolorise the blue starch iodine complex. Select the correct statement(s) :

- (A) The mass of AgNO_3 initially taken is 680 g
- (B) The mass of AgNO_3 initially taken is 850 kg
- (C) Fe required is 2000 mole
- (D) Fe required is 2500 mole

6. Which of the following statement(s) is/are true for ideal gas ?
- (A) $(-w_{\text{rev}}) - (-w_{\text{irr}}) > 0$ for isothermal expansion.
- (B) $w_{\text{rev}} > w_{\text{irr}}$ for isothermal compression
- (C) Heat rejected in isothermal process may be greater in $(n-1)^{\text{th}}$ stage as compared to n^{th} stage work.
- (D) For expansion and compression both work done by the system is maximum for reversible process.
7. Reaction with acidified permanganate solution of $(\text{CH}_3)_2\text{C}=\text{CH}-\text{CH}_2-\text{COOH}$ gives (A) and (B) compounds. A gives positive Iodoform test. Choose correct statement(s) of compound (A) and (B)
- (A) Compound (A) gives 2° -alcohol by LiAlH_4 in T.H.F.
- (B) Compound (B) is heated to give acetic acid
- (C) Compound (A) gives Iodoform test
- (D) Compound (B) is stronger acid than oxalic acid
8. Predict product (A), (B) (C) and (D)?



- (A) A is 
- (B) D is 
- (C) B is 
- (D) A is 



Choose incorrect statement for compound (X) and (Y)

- (A) Compound (X) is chiral
 - (B) Compound (Y) is 2-oxo-cyclopentanecarboxylic acid
 - (C) Compound (Y) is react with hydroxyl amine to give two geometrical isomers
 - (D) Degree of unsaturation of (X) is 3
10. Which of the following statement is/are correct towards electrophilic substitution reaction when given groups are attached to benzene ring-
- (A) $-\text{NO}_2$ is a metadirecting and deactivating group
 - (B) $-\text{Cl}$ is a metadirecting and deactivating group
 - (C) $-\text{OMe}$ is an ortho-paradirecting and activating group
 - (D) $-\text{CH}_3$ is an ortho-paradirecting and activating group

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

In qualitative analysis, the observations of each test must be carefully observed along with pH conditions and temperature conditions of the solution.

- How many of the following steps/observations is/are wrong regarding brown ring test of NO_3^- ?
 - Ferrous sulphate can be allowed to stand before mixing it with the solution
 - A dark brown ring of nitroso ferrous solution is formed at the junction of the solution
 - In 1 ml of aqueous solution of salt, 2 mL of dil H_2SO_4 needs to be added
 - In case of presence of NO_2^- along with NO_3^- , use of sulphamic acid can help with the test.
 - The complex formed is coloured due to charge transfer spectrum
 - In case of formation of white ppt. while mixing conc. H_2SO_4 , the test can be carried out without any measures.
- For the complex formed in sodium nitroprusside test of sulphide ion,
 Difference between magnitude of charge on product complex ion and reactant complex ion
 = _____
 Number of linkage isomers if ambidentate behaviour of CN^- is not considered

Paragraph for Questions 3 and 4

Consider the following elementary reaction $\text{X}_2(\text{g}) \xrightleftharpoons[K_b = 4\text{sec}^{-1}\text{torr}^{-1}]{K_f = 4\text{sec}^{-1}} 2\text{X}(\text{g})$ at 2000°C

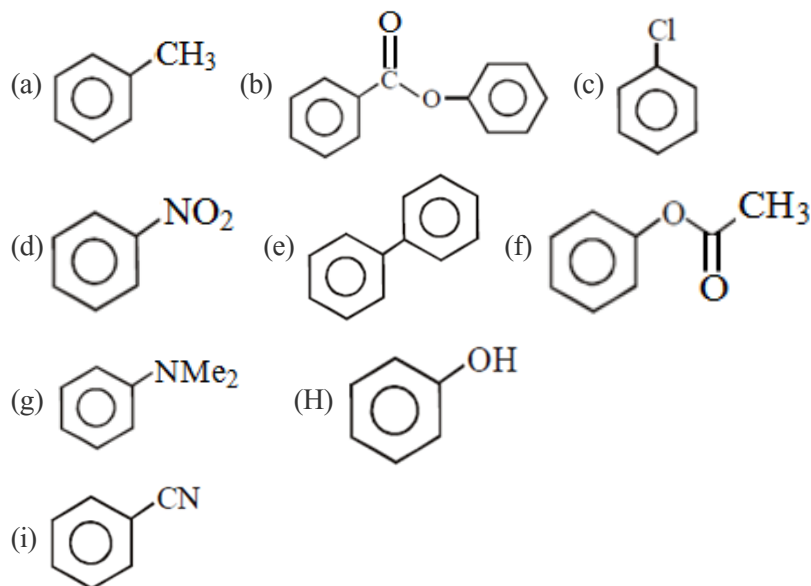
and at 2 torr equilibrium pressure. The composition of this equilibrium mixture is determined by measuring the rate of effusion through a pin hole. It is observed that the mixture effuses 1.5 times slower as helium effuses under identical conditions.

- Calculate molar mass of the gas X_2 in gram per mol.
- If above elementary reaction is assumed to be irreversible then calculate time in sec in which amount of reactant reacted is exactly equal to the amount of reactant reacted to reach equilibrium when reaction was occurring reversibly. $\left(\text{Given : } \ln \frac{3}{2} = 0.4\right)$

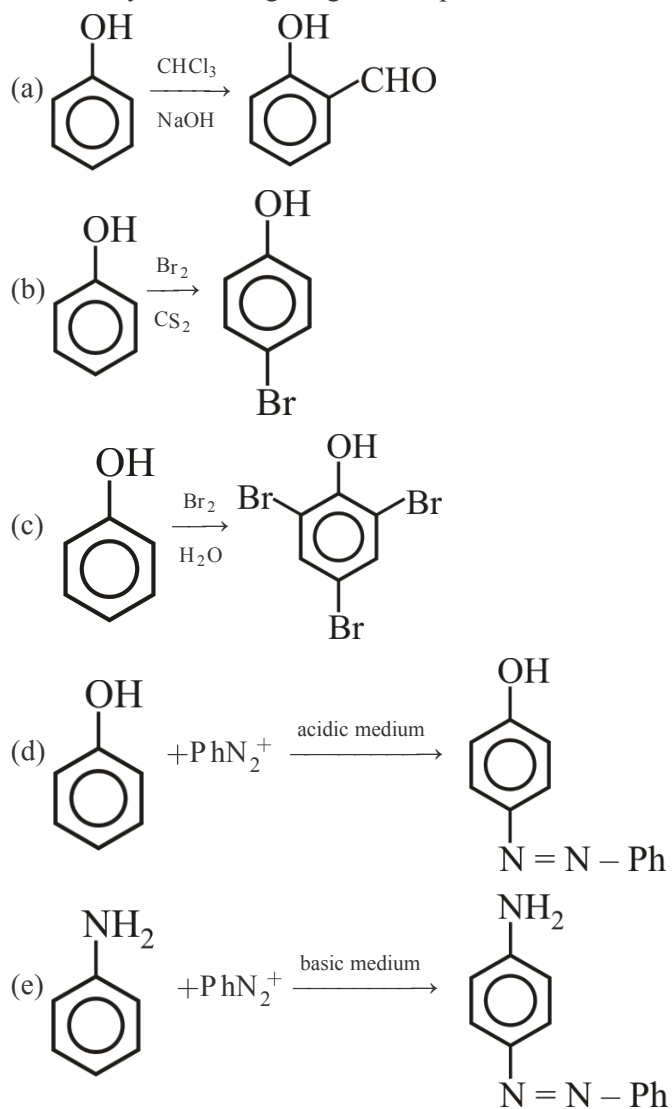
Paragraph for Questions 5 and 6

Activating groups in benzene are more reactive and deactivating groups in benzene are less reactive.

5. How many of the following compounds will show higher rate of electrophilic substitution reaction than benzene -



6. How many reactions giving correct product?



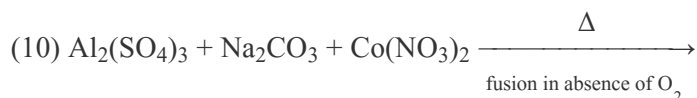
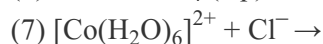
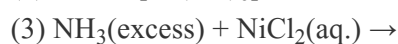
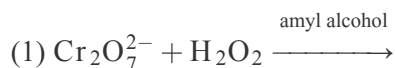
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. How many of the following processes involves blue colouration?



8. Let 'A' and 'B' are two completely miscible liquids. $P_A^\circ = 90$ torr and $P_B^\circ = 40$ torr. In a flask 8 moles of 'A' is mixed with 4 moles of 'B'. However as soon as B is added 'A' starts dissociating as two different liquids 'C' and 'D' as $A \rightarrow C + D$.

Both 'C' and 'D' are also completely miscible with each other and with 'A' and 'B'. Given $P_C^\circ = 50$ torr and $P_D^\circ = 40$ torr.

Assume dissociation of 'A' follows 1st order kinetics with half life equal to 50 min. After 100 min 2 moles of a non-volatile non-electrolyte solute 'S' is added in solution which immediately arrests the dissociation of A completely.

Calculate final vapour pressure of the solution in torr.

9. Alpha particles capture free moving electrons having wavelength $5\text{\AA}$ and form excited H-like species. In this excited state, the circumference of orbit is 2 times the wavelength of electron. If photons emitted in this process fall on a metal surface of work function 10.6 eV. Find the maximum possible kinetic energy (in eV) of photoelectrons.

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Let $f : [1, \infty) \rightarrow (-\infty, -1]$ be a continuous and differentiable function such that $f(x) = x - 2x^2 + \int_1^x \frac{f(t)}{x} dt$ for all $x \in [1, \infty)$. Then, which of the following statement(s) is(are) True ?
 (A) The curve $y = f(x)$ passes through the point $(3, -24)$
 (B) Function $f(x)$ is an onto function
 (C) Function $f(x)$ is an into function
 (D) The area of the region $\{(x, y) \in [1, 2] \times \mathbb{R} : f(x) < y < -x\}$ is $\frac{7}{2}$
2. Let A be a 3×3 matrix. If every column vector X satisfies $X^T A X = 0$ and $a_{12} = 2008$, $a_{13} = 1010$, $a_{23} = -2012$. Then the value of $a_{21} + a_{31} + a_{32}$ is (where X^T represents transpose of matrix X)
 (A) -6 (B) 1006 (C) -1006 (D) 0
3. The solution of the differential equation $\frac{dy}{dx} + x(x + y) = x^3(x + y)^3 - 1$ is : (Where c is arbitrary constant)
 (A) $\frac{1}{x + y} = x^2 + 1 + ce^x$ (B) $\frac{1}{(x + y)^2} = x^2 + 1 + ce^{x^2}$
 (C) $\frac{1}{(x + y)^2} = x + 1 + ce^x$ (D) $\frac{1}{x + y} = x + 1 + ce^{x^2}$
4. Consider a real valued continuous function f such that $f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + t f(t)) dx$. If M and m are maximum and minimum value of the function f , then
 (A) $\frac{M}{m} = 4$ (B) $M - m = 2\pi + 1$
 (C) $M + m = 4(\pi + 1)$ (D) $Mm = 2(\pi^2 + 1)$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. If A and B are two matrices such that $AB = A$ and $BA = B$, then

(A) $(A^3 - B^3)^5 = (A^5 - B^5)^3$

(B) $(A^4 - B^4)^3 = A^{12} - B^{12}$

(C) $(A - B)$ is idempotent

(D) $(A^{10} - B^{10})$ is nilpotent

6. Let $\int \frac{dx}{x^2 + ax + 1} = f(g(x)) + C$, where C is constant of integration, then

(A) $f(x)$ is inverse trigonometric function for $|a| < 2$

(B) $f(x)$ is logarithmic function for $|a| > 2$

(C) $g(x)$ is quadratic function for $|a| < 2$

(D) $g(x)$ is rational function for $|a| > 2$

7. Let A is a non-singular matrix of order $n \times n$ such that $3ABA^{-1} + A = 2A^{-1}BA$, then which is/are correct (where $|A| = \det(A)$) -
- (A) A and B both are identity matrix
- (B) $|A + B| = 0$
- (C) $|ABA^{-1} - A^{-1}BA| = 0$
- (D) $A + B$ is not a singular matrix.
8. Let $f(x)$ be a non-constant, twice differentiable function defined on $\mathbb{R}$ such that $y = f(x)$ is symmetric about line $x = 1$ and $f(-1) = f'\left(\frac{1}{4}\right) = f'\left(\frac{1}{2}\right) = 0$, then which of following is/are correct statement(s) ?
- (A) $f''(x) = 0$ has atleast four roots in the interval $(0, 2)$
- (B) $\int_{-2}^2 (x^5 + x^3) f(1+x) dx < \int_0^2 \frac{1}{1+2^{f(x)}} dx$
- (C) there exist at least one $C \in \left(\frac{3}{2}, \frac{7}{4}\right)$ such that $f'(C) + Cf''(C) = 0$
- (D) for some $C \in \left(1, \frac{3}{2}\right)$, $f'(C) = Cf''(C)$
9. Let f be twice differentiable real valued function satisfying $f'(x) \neq 0$, $f(x) + f''(x) = -xg(x)f'(x)$ where $g(x) > 0 \forall x > 0$. If $f(0) = -3$ and $f'(0) = 4$ then
- (A) $|f(x)| \leq 5 \forall x > 0$
- (B) $(f(x))^2 + (f'(x))^2$ is decreasing $\forall x > 0$.
- (C) $|f(x)| > 5 \forall x > 0$
- (D) $(f(x))^2 + (f'(x))^2$ is increasing $\forall x > 0$.
10. Given $f(x) - 2 \cdot \int_0^{\frac{\pi}{4}} \left(\frac{\sin^2 t}{\cos^5 t} \right) f(t) \cos t dt = \frac{\sin^2 x}{\cos^5 x}$
- (A) If $f : \mathbb{R} - \left\{ (2n+1)\frac{\pi}{2} \right\} \rightarrow \mathbb{R}$, then f is one-one ; $n \in \mathbb{Z}$
- (B) $f(x)$ is periodic function with fundamental period π
- (C) If $f : \mathbb{R} - \left\{ (2n+1)\frac{\pi}{2} \right\} \rightarrow \mathbb{R}$, then f is onto ; $n \in \mathbb{Z}$
- (D) $f\left(\frac{\pi}{3}\right) = 72$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Let $A = [a_{ij}]_{3 \times 3}$ be upper triangular matrix of order 3 and its characteristic equation is $A^3 + \alpha A^2 + \beta A - I = 0$, where $a_{ij} \in \{-1, 0, 1\}$. (where α, β are scalars)

1. Sum of all the possible values of α is -
2. If $\alpha = -3$, then value of β is -

Paragraph for Questions 3 and 4

Let $f : [0, \infty) \rightarrow [2, \infty)$ be a derivable function which is also surjective such that $f(0) = 2$ and satisfying $f^2(x) - f^2(y) = 3f(x) - 3f(y) + \sqrt{x} - \sqrt{y}$

3. The value of $\lim_{x \rightarrow 0^+} \left(\frac{2f(x) - 3 - e^{\sqrt{x}}}{\sqrt{x}} \right)$ equals
4. If g is the inverse function of f then $\frac{d}{dx} \left(\frac{4}{g(x)} \right)$ at $x = 3$ is equal to

Paragraph for Questions 5 and 6

Consider a function $y = f(x) = \sin x + \cos x + |\sin x - \cos x|$, $x \in [0, 2\pi]$.

5. The value of $\lim_{h \rightarrow 0} \frac{(f(\pi + h))^2 - (f(\pi - h))^2}{h}$ is -
6. Number of points in $[0, 2\pi]$ where $f(x)$ is non differentiable is -

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Let $A = \begin{bmatrix} \frac{1}{2} & \frac{3}{4} \\ 1 & -\frac{1}{2} \end{bmatrix}$ then the value of $\text{tr}(A) + \text{tr}(A^2) + \text{tr}(A^3) + \dots + \text{tr}(A^{100})$, is

(Where $\text{tr}(A)$ denotes trace of matrix A)

8. If the eccentricity of curve for which tangent at point P intersects the y axis at M above x -axis such that origin is equidistant from M and point of tangency is e then value of $6e^2$ is ?
9. If $f(x) = e^{(x+1)^n}$; ($n \in \mathbb{N}$) and $f''(1) = 67ke^k$ then value of $(n+k)$ is -

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work